Personal Learning Journey

IT3091 Machine Learning - Group 2026-AI-05

DataCo Smart Supply Chain: late-delivery risk

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Area of responsibility: Exploratory data analysis, evaluation and recommendation

My role

I contributed to the exploratory data analysis and then owned the evaluation, the model comparison and the final recommendation. My job was to decide what our results actually mean and what we could honestly claim.

What I did

On the EDA side I worked on the time-based analysis, including the monthly stability of the late rate and the delay patterns by hour of day and day of week.

For the evaluation I compared all seven models on six metrics, checked the cross-validation logic, ran the threshold analysis, produced the final test evaluation and wrote the recommendation and limitations.

What I learned

The most important thing I learned is to ask whether a difference is real before reporting it. Our top three models are separated by 0.0022 ROC-AUC, but a single model varies by about 0.0030 between folds. The difference between the models is smaller than the variation within one model, so they are statistically tied. We chose Random Forest for its stability and interpretability rather than claiming it won.

I also learned that a default decision threshold is a choice, not a law. At 0.50 we catch 60% of late orders. At 0.40 we catch 79%. Since missing a late order costs a customer while a false alarm costs one shipping upgrade, 0.50 was the wrong trade-off for this business problem.

Finally I learned that the model was not the most valuable thing we produced. The finding that First Class is late 100% of the time, because it promises one day and always takes two, is worth far more to DataCo than a 0.78 classifier.

What I found difficult

The threshold analysis caught me out. I started by optimising F1, which is the standard thing to do, and it picked a threshold that flags 97% of all orders. That is the always predict late model in disguise, with a respectable looking score. I had to work out why: the predicted probabilities form a few separate clusters because Shipping Mode dominates, so there is no smooth trade-off to tune.

Writing the limitations honestly was also uncomfortable. It is tempting to present 0.78 as a good result rather than explain that the ceiling is in the data.

A mistake I caught

Our cross-validation used a time-series split, which cuts the data at a row position. Because the rows of one order sit next to each other, a fold boundary could land inside an order and leak its label. I measured it and found two orders affected. It was a small number, but it was exactly the leak we had rejected random splitting for, so we wrote a splitter that removes those rows.

What I would do differently

I would decide the evaluation metrics and the business cost of each error type before any model was trained. We arrived at recall mattering more than precision late in the project, and that decision should have shaped the work from the start.

On using AI

I used Claude for the evaluation code and for drafting the report. It was helpful, but it predicted that the test score would fall below validation and it actually rose by 0.0103, and it once titled a chart claiming three methods agreed when one clearly disagreed. Checking the output against the data is what caught both.